

Dental Health Knowledge Base

Clinical Reference Document — v1.0

This document is the authoritative knowledge source for the Dental Health Chatbot. The chatbot must answer questions solely based on the content of this document. Any question outside the scope of this document should receive a polite refusal.

1. Common Dental Procedures

1.1 Root Canal Treatment

A root canal is a procedure to treat infection or inflammation inside a tooth. The dentist removes the infected or inflamed pulp (nerve and blood vessels), cleans and shapes the root canals, then fills and seals them with a material called gutta-percha. A crown is usually placed afterward to protect and restore the tooth's full function.

Root canals are necessary when decay reaches the pulp, when a tooth is cracked deeply, or when repeated dental procedures have weakened the tooth. The procedure typically takes one to two appointments and is performed under local anaesthesia.

1.2 Tooth Extraction

Tooth extraction is the removal of a tooth from its socket in the jawbone. It is performed when a tooth is too damaged, decayed, or crowded to be saved. Simple extractions are performed on teeth that are visible above the gumline. Surgical extractions are required for teeth that are impacted, broken at the gumline, or have not fully erupted.

After an extraction, patients are advised to avoid smoking, rinsing vigorously, or using straws for at least 24 hours to protect the blood clot that forms in the socket.

1.3 Dental Fillings

A dental filling is used to restore a tooth damaged by decay back to its normal shape and function. The dentist removes the decayed material, cleans the affected area, and fills the cavity with one of the following materials: composite resin (tooth-coloured), amalgam (silver), gold, or porcelain. Composite resin is the most common choice for visible teeth due to its natural appearance.

1.4 Scaling and Root Planing

Scaling and root planing is a non-surgical deep-cleaning procedure used to treat gum disease. Scaling removes plaque and tartar deposits from tooth surfaces and from below the gumline. Root planing smooths

the tooth root surfaces to help the gums reattach to the teeth and to prevent bacteria from re-accumulating. Local anaesthesia is often used during this procedure.

1.5 Dental Implants

A dental implant is a titanium post surgically inserted into the jawbone to replace a missing tooth root. After a healing period of three to six months (osseointegration), an abutment and a custom crown are attached to the implant to look and function like a natural tooth. Implants are considered the gold standard for replacing missing teeth.

2. Dental Crowns

A dental crown is a tooth-shaped cap that is placed over a damaged, decayed, or weakened tooth to restore its shape, size, strength, and appearance. Crowns are cemented into place and cover the entire visible portion of the tooth above the gumline.

When is a Crown Needed?

A dentist may recommend a crown in the following situations: after a root canal treatment to protect the restored tooth; for a severely decayed tooth where a filling is not sufficient; for a cracked or fractured tooth; to anchor a dental bridge in place; or to cover a misshapen or severely discoloured tooth.

Crown Materials

Material	Best For	Notes
Porcelain / Ceramic	Front teeth	Best colour match; most natural appearance
Metal alloy (gold/silver)	Back molars	Very durable; less aesthetically pleasing
Porcelain-fused-to-metal	Any tooth	Strong with a natural look; some metal may show
Zirconia	Any tooth	Very strong and tooth-coloured; modern choice

3. Gum Disease (Periodontal Disease)

3.1 Gingivitis

Gingivitis is the earliest and mildest stage of gum disease. It is characterised by red, swollen, and bleeding gums, particularly when brushing or flossing. Gingivitis is caused by plaque buildup along the gumline and is fully reversible with proper oral hygiene and professional dental cleaning. At this stage, no bone or tissue damage has occurred.

3.2 Periodontitis

If gingivitis is left untreated, it can progress to periodontitis. In periodontitis, the inner layer of the gum and bone pull away from the teeth, forming pockets that collect debris and become infected. The body's immune response and bacterial toxins begin to break down the bone and connective tissue that hold teeth in place. Without treatment, the teeth may become loose and need to be removed.

3.3 Signs and Symptoms

- Bleeding gums during or after brushing
- Red, swollen, or tender gums
- Persistent bad breath (halitosis)
- Receding gums (teeth appear longer than normal)
- Deep pockets between teeth and gums
- Loose or shifting teeth
- Pain when chewing

3.4 Risk Factors

Common risk factors for gum disease include: smoking or tobacco use, diabetes, certain medications that reduce saliva flow, hormonal changes (pregnancy, menopause), genetic susceptibility, and poor oral hygiene habits.

4. Orthodontics

4.1 Braces

Braces are orthodontic devices used to correct misaligned teeth and jaws. They consist of brackets bonded to teeth and connected by archwires. The wire applies constant, gentle pressure over time to gradually move teeth into the correct position. Treatment typically lasts between 18 months and 3 years depending on the severity of the case.

4.2 Clear Aligners

Clear aligners such as Invisalign are a series of transparent, removable plastic trays used as an alternative to traditional braces. Each tray is worn for one to two weeks before being replaced with the next in the series, progressively moving the teeth. Aligners should be worn 20 to 22 hours per day and removed only for eating, drinking, brushing, and flossing.

4.3 Retainers

Retainers are custom-made devices worn after orthodontic treatment to maintain teeth in their corrected positions and prevent them from drifting back. Retainers may be removable or fixed (bonded behind the teeth). Patients are typically advised to wear retainers full-time initially and then only at night long-term.

5. Preventive Dental Care

5.1 Dental Sealants

Dental sealants are thin, protective plastic coatings applied to the chewing surfaces of back teeth (molars and premolars) to prevent tooth decay. The sealant bonds into the grooves of the teeth, forming a shield over the enamel. Sealants are most commonly recommended for children but adults without decay or fillings in their molars can also benefit.

5.2 Fluoride Treatment

Fluoride is a naturally occurring mineral that strengthens tooth enamel and helps prevent cavities by making teeth more resistant to acid attacks from plaque bacteria and sugars. In-office fluoride treatments are applied as a gel, foam, or varnish during a dental visit and take only a few minutes. Patients are advised not to eat or drink for 30 minutes after the application.

5.3 Professional Dental Cleaning (Prophylaxis)

A professional dental cleaning removes plaque and tartar that regular brushing and flossing cannot reach. During the cleaning, a dental hygienist scales the teeth, polishes them, and may apply fluoride. A cleaning is recommended every six months for most patients, though patients with gum disease may need more frequent visits.

5.4 Dental X-rays

Dental X-rays (radiographs) are an essential diagnostic tool that allow dentists to detect problems not visible during a regular oral examination, including cavities between teeth, bone loss, infections, impacted teeth, and abnormalities. Bitewing X-rays, the most common type, are typically taken once a year. Full-mouth X-rays are taken every three to five years.

6. Cosmetic Dentistry

6.1 Teeth Whitening

Teeth whitening is a cosmetic procedure that uses peroxide-based bleaching agents to lighten the colour of teeth. In-office whitening, performed by a dentist, produces results in a single appointment using a high-concentration bleaching agent, sometimes activated by a light or laser. At-home whitening uses custom-fitted trays and a lower-concentration gel worn for one to two hours a day or overnight for two to four weeks.

Whitening is not permanent. Results typically last from six months to two years depending on diet, smoking habits, and oral hygiene. Whitening does not work on crowns, veneers, or fillings.

6.2 Dental Veneers

Dental veneers are thin shells of porcelain or composite resin that are custom-made and bonded to the front surface of teeth to improve their appearance. Veneers can correct the look of stained, chipped, broken, or slightly misaligned teeth. Porcelain veneers are more durable and resist stains better than composite veneers. The procedure is generally irreversible because a small amount of enamel is removed from the tooth surface.

7. Emergency Dental Care

7.1 Toothache

A toothache may be caused by tooth decay, a cracked tooth, an abscess, gum disease, or an exposed tooth root. For temporary relief, over-the-counter pain relievers and clove oil applied to the affected area may help. A dentist should be seen as soon as possible to identify and treat the underlying cause.

7.2 Knocked-Out Tooth (Avulsion)

If a permanent tooth is knocked out, time is critical. The tooth should be picked up by the crown (not the root), rinsed gently with water if dirty (do not scrub), and either re-inserted into the socket immediately or kept moist in milk or a saline solution. The patient should see a dentist within 30 minutes for the best chance of saving the tooth.

7.3 Dental Abscess

A dental abscess is a pocket of pus caused by a bacterial infection. It can occur at the tip of the tooth root (periapical abscess) or in the gums (periodontal abscess). Symptoms include severe, persistent toothache, sensitivity to hot and cold, fever, swelling in the face or cheek, and swollen lymph nodes. A dental abscess is a serious condition that requires prompt dental treatment — it will not resolve on its own.

7.4 Broken or Chipped Tooth

If a tooth is broken or chipped, any sharp edges should be covered with dental wax or sugar-free gum to protect the tongue and cheeks. Rinsing with warm salt water helps reduce the risk of infection. The dentist will assess the damage and may recommend bonding, a crown, or extraction depending on the severity.

This document is intended solely as the knowledge source for the Dental Health Chatbot. It does not constitute professional medical advice. Patients should always consult a qualified dental professional for diagnosis and treatment.